



LLM Evaluation Metrics: Cheat Sheet

Faithfulness, relevancy, BLEU, ROUGE, and perplexity:
what each one actually tells you.



1. FAITHFULNESS

Grounded in Context



2. RELEVANCY

Answer & Context



3. BLEU / ROUGE

N-Gram Overlap



4. PERPLEXITY

Model Surprise



5. PERPLEXITY

Model Fluency





OVERVIEW

Not All Metrics Measure The Same Thing

Picking the wrong metric gives precise numbers that mean nothing useful



The Core Evaluation Challenge

Engineers often grab whatever metric appears first in a tutorial. However, these five core metrics belong to distinct mathematical and practical categories.

RAG Grounding

Faithfulness /
Relevancy



N-Gram Match

BLEU / ROUGE



Model Fluency

Perplexity



Three Distinct Mathematical Categories



Semantic Grounding (RAG): Evaluates factual support against context without requiring exact word match.



Reference Overlap (N-Grams): Measures exact phrase or subsequence match against human reference strings.



Next-Token Probability: Measures model predictability and fluency regardless of factual truth.



Design Rule: Never evaluate RAG grounding with translation metrics, or model fluency with recall scores.



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Faithfulness: **Grounded in Context**

Is every claim supported by the retrieved context?

What It Actually Measures

Measures whether every claim in an answer is supported by the retrieved chunks. It does NOT measure absolute world correctness: only factual consistency with retrieved context.

```
from ragas import evaluate
from ragas.metrics import faithfulness
from datasets import Dataset

data = Dataset.from_dict({
    "question": ["Refund window?"],
    "answer": ["30 days for physical items."],
    "contexts": [["Returns allowed within 30 days."]]
})
scores = evaluate(data, metrics=[faithfulness]) # {'faithfulness': 1.0}
```

Generated Answer
split into claims



NLI Entailment Judge
claim vs context check



Score 0.0 to 1.0

- ✓ **Interpretation (0 to 1):** 1.0 means every claim is backed by context. Lower scores indicate hallucinated or ungrounded claims.



Relevancy: Answer & Context

Addressing the question and retrieving high-signal chunks

Two Distinct Dimensions

1. **Answer Relevancy:** Does the generated response directly answer what was asked (not just correct, but on-topic)?
2. **Context Precision:** Did retrieval pull back high-signal chunks useful for answering, or mostly noisy text?

```
from ragas.metrics import answer_relevancy, context_precision

scores = evaluate(
    eval_data,
    metrics=[answer_relevancy, context_precision]
)
# Result: {'answer_relevancy': 0.92, 'context_precision': 0.75}
```

User Query
intent check



Semantic Alignment
embedding cosine similarity



Score 0.0
to 1.0



Diagnostic Signal: Low answer relevancy = off-topic generation. Low context precision = retriever pulling too much noise.



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BLEU: N-Gram Word Overlap

Bilingual Evaluation Understudy against reference text



What It Actually Measures

Measures exact 1-gram to 4-gram word overlap precision between generated text and reference text, multiplied by a brevity penalty for short outputs.

```
from nltk.translate.bleu_score import sentence_bleu

ref = [["the", "cat", "sat", "on", "the", "mat"]]
cand = ["the", "cat", "is", "on", "the", "mat"]

score = sentence_bleu(ref, cand) # e.g., 0.55 (penalized for word change)
```

N-Gram Matching

1-gram through 4-gram



Brevity Penalty

penalize overly short text



**BLEU
Score**



Critical Limitation: BLEU penalizes valid paraphrases heavily because it only rewards exact wording match. Bad fit for open-ended RAG.



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ROUGE: Content Recall

Recall-Oriented Understudy for Gisting Evaluation

Recall vs. Precision Focus

Measures how much of the reference text's content is captured in the generated output.

ROUGE-1: Unigram word recall | **ROUGE-2:** Bigram recall

ROUGE-L: Longest Common Subsequence (captures sentence flow).

```
from rouge_score import rouge_scorer

scorer = rouge_scorer.RougeScorer(["rouge1", "rougeL"],
use_stemmer=True)
scores = scorer.score(
    "the cat sat on the mat",
    "the cat is sitting on the mat"
)
# Returns precision, recall, and F1 for each variant (0.0 to 1.0)
```

Reference Facts
key content to cover



LCS / N-Gram Recall
check coverage proportion



ROUGE F1

- ✓ **Best For:** Summarization tasks where you must verify whether key source facts made it into the generated summary.



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Perplexity: Model Surprise

Predicting token probability, NOT factual truth



What It Actually Measures

Perplexity measures how "surprised" a model is by a sequence of text (exponential of negative log-likelihood). Lower score = higher token predictability and fluency.

```
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer

def calc_perplexity(text):
    inputs = tokenizer(text, return_tensors="pt")
    with torch.no_grad():
        loss = model(**inputs, labels=inputs["input_ids"]).loss
    return torch.exp(loss).item()
# Lower perplexity = text looks more "expected" by model weights
```

Input Token Sequence
evaluate probabilities



exp(Loss)
negative log-likelihood



Lower = Better



Critical Warning: A model can produce extremely low-perplexity (fluent) hallucinations! Perplexity measures fluency, not truth.



All 5 Metrics: Side by Side

Summary matrix of focus area, range, and reference needs

METRIC	WHAT IT MEASURES	BEST FOR	RANGE	REFERENCE?
Faithfulness	Grounded in retrieved context?	RAG grounding	0 to 1	No (Context)
Relevancy	Addresses question & high-signal?	RAG precision	0 to 1	No
BLEU	Exact n-gram word overlap	Translation	0 to 1	Yes
ROUGE	Content recall & subsequence	Summarization	0 to 1	Yes
Perplexity	Model surprise & fluency	Base LLM eval	0 to ∞	No

Golden Rule of LLM Benchmarking

Reference-free metrics (Faithfulness / Relevancy) use semantic LLM judges against context. Reference-based metrics (BLEU / ROUGE) require curated human ground-truth strings.


Engineering Tip: Modern RAG evaluation pipelines rely almost entirely on Faithfulness + Relevancy rather than n-gram overlap.



Matching Metrics To **Your Application**

Pick the metric that aligns with your real engineering goal

Building a RAG System?

Use **Faithfulness + Answer Relevancy + Context Precision**. These directly inspect grounding and retrieval accuracy.

Building a Summarizer?

Use **ROUGE-1 and ROUGE-L** to verify key facts and sentence sequence retention against human summaries.

Fine-Tuning or Comparing Base LLMs?

Use **Perplexity and Evaluation Loss** to see which checkpoint fits your target domain text distribution best.

What NOT To Do

Never use **BLEU or ROUGE** alone to judge open-ended QA or agentic assistants: paraphrased correct answers get penalized unfairly.

 **Actionable Rule:** Align your evaluation metric with how your end-users judge success.



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ADVANCED EVALUATION

Beyond These 5: **Extended Toolkit**

Complementary metrics for production-grade evaluation



Exact Match (EM) & F1

Standard for extractive QA benchmarks (SQuAD style). Checks whether the answer span matches exactly token-for-token.



BERTScore

Uses contextual embedding similarity between candidate and reference tokens, correctly rewarding valid semantic paraphrasing.



Context Recall

Measures what proportion of all relevant ground-truth facts were successfully retrieved by your vector database.



Human Evaluation (Gold Standard)

Essential for evaluating tone, empathy, and nuanced reasoning. Always calibrate automated judges periodically against human scores.



Production Stack: Combine embedding metrics (BERTScore) with LLM-as-judge (Ragas) and periodic human review.



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Common Evaluation Pitfalls

Five mistakes that invalidate your metric dashboard

- 1. N-Grams on Open-Ended QA:** Using BLEU/ROUGE on conversational answers punishes excellent paraphrases.
- 2. Perplexity as Quality:** Assuming low perplexity means factual truth. It only means high token fluency.
- 3. Mean Without Distribution:** A 0.88 average faithfulness can hide a 10% severe hallucination tail.
- 4. Blind LLM-as-Judge:** Trusting automated judges without spot-checking human correlation.
- 5. Mismatched Preprocessing:** Comparing scores across different tokenizers or reference formatting.

Automated Score

e.g. 0.92 mean



Inspect Outliers

check lowest 5% tails



Reliable System



Best Practice: Always monitor your p95 and p99 failure tails rather than trusting average metric values.



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Pick The Metric That **Matches Your Task**

Use the right tool, or your metric score won't mean what you think

✓ **Faithfulness**

RAG Grounding & Hallucination Check

🎯 **Relevancy**

Answer Topic & Context Signal Precision

” **BLEU**

Machine Translation N-Gram Match

✂ **ROUGE**

Summarization Content Recall

🗣 **Perplexity**

Base Model Predictability & Fluency

📌 **Save & Reference**

Save this as your LLM evaluation reference sheet. Follow for more practical AI engineering breakdowns.



Final Takeaway: Never evaluate grounded RAG systems with pure n-gram overlap metrics.



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